

CS 577 — Fall 2025 — Homework 3

Problem 1 — [10] point(s). In lecture, we designed `im2col` to convert patches shown in Figure 1-left to row vectors. For this problem, modify the `im2col` function to convert “patches” in Figure 1-right to row vectors.

The python notebook for this problem is in `prob1-im2col.ipynb`.

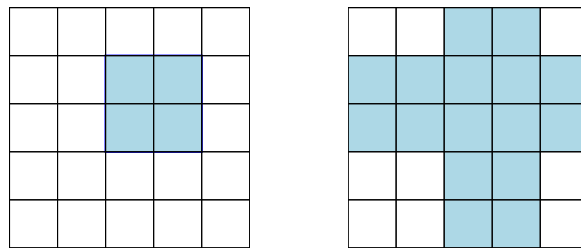


Figure 1: Left: standard patches. Right: modified “patches”.

More precisely, the modified patch should contain the original K -by- K rectangular patch, as well as extension of the rectangle vertically and horizontally to the boundaries of the image.

Problem 2 — [10] point(s). In lecture, we implemented a 1-layer recurrent neural network (with ReLU activation) to predict the “max” of a sequence. Implement a 2-layer recurrent neural network (with ReLU activation) as in below that predicts the **top-2 largest** elements of an input sequence.

The python notebook for this problem is in `prob2-RNN.ipynb`.

You should implement the `forward` function, as well as the three lines in the main training loop below the “YOUR CODE HERE ” comment.

```

+-----+
|         |
h_2(t-1) >-->|Wrec2|>--> h_2(t)
+-----+
      Λ
      | Win_2
      |
+-----+
|         | h_1(t)
+-----+ |
|         | |
h_1(t-1) >-->|Wrec1|>--> h_1(t)
+-----+
      Λ
      | Win_1
      |
x(t)

```

Note:
/ these are the same tensors
| just used in more than one place

There is a code block for sanity check at the bottom of the page. Hint: you can refer to `lec08-in-class-ex3-vrec-max-SOLUTION.ipynb` in the Lecture 8 announcement post for vectorized 1-layer RNN.